

Econometric game 2025

Anastasija Tetereva, Onno Kleen, Alla Petukhina

11 April 2025

1 Case B

- Visualize the time series data that will be used for prediction, along with their relationships to key factors such as weather features, electricity price features, electricity generation and consumption forecasts, and natural fuel market features. Overlay or create side-by-side charts for the related features. Create scatter plots or correlation matrices to show the relationship between the predicted time series and the related features. Highlight any visible trends or correlations between the time series and related features. Offer insights into which factors seem to have the most significant impact on the target prediction.
- The primary objective is to develop a day-ahead, a week-ahead, a month-ahead predictive model to forecast the total load of redispatch within the German electricity network, in particular, for TenneT DE network. This model should operate on an hourly basis in Central European Time (CET) and provide forecasts for
 - **the expected total load,**
- In designing an adequate day-ahead forecast scenario, what training and forecasting routines did you adopt? For training purposes, consider using data from the period spanning January 1, 2020, to June 30, 2022. Use data from July 1, 2022, to December 31, 2022, to validate your model. Apply data from 2023 to evaluate the model's performance comprehensively.
- What specific steps would you take to ensure robust validation?
- What key metrics or indicators would you focus on in this evaluation?
- Describe your approach to backtesting. Would you implement an expanding or a rolling window technique, and what are the advantages and potential limitations of your chosen method?
- How would you assess the model's predictive accuracy across different forecast horizons? What insights could you provide based on this evaluation? Provide a single-execution forecast for the entirety of 2023. What are the

key steps involved in this process, and how would you present the results effectively?

- Using visualizations, identify and discuss specific periods in 2023 when your model performed well and when it struggled. What underlying factors might explain these variations in performance, and how could the model be further refined based on these observations?
- Provide policy recommendations based on your results.
- If there are several observations with the same timestamp, group them by hour.

2 Instructions for structuring the report

Your report should be clear, structured, and well-argued, demonstrating both methodological rigor and economic intuition. Below is the recommended structure:

1. Introduction
Briefly introduce the problem and its economic relevance. State your research question and objectives. Outline the structure of the report.
2. Data Description
Describe the dataset used, including sources, sample size, and time period. Explain key variables, transformations, and any data cleaning steps. Provide summary statistics and visualizations where relevant.
3. Methodology
Clearly explain the econometric model(s) applied. Justify the choice of estimation techniques. Discuss any assumptions and potential econometric issues.
4. Empirical Results
Present key estimation results in tables and figures. Interpret the coefficients and discuss statistical significance. Perform robustness checks and discuss potential limitations.
5. Discussion and Economic Interpretation
Relate your findings to economic theory and real-world implications. Compare results with existing literature. Discuss policy recommendations or business implications if relevant.
6. Conclusion
Summarize key findings concisely. Highlight limitations and potential extensions for future research.
7. References
List all academic papers, datasets, and other sources cited. Use a consistent citation format.

8. Appendix (if necessary)
Include additional tables, robustness tests, or technical derivations that support the main text.
9. The total volume of the document including Appendix should not exceed 20 pages. Be mindful of page formatting: use standard font size (e.g., 12 pt), appropriate margins (e.g., 1 inch), and spacing (e.g., 1.5 or 2x line spacing).

3 Instructions for structuring the presentation

1. Introduction
Brief introduction (1-2 lines) to the research or topic.
Goal: Set the context and the research question for your audience.
2. Methodology
Outline the methodology used in the analysis or research.
Keep it simple but informative.
3. Summary statistics
Present key summary statistics.
Brief commentary on what these statistics reveal about the data.
4. Main results
Present the main findings or results of your analysis.
Be concise and focus on the most important outcomes.
5. Interpretation and policy recommendations
Provide a brief interpretation of the results. What do the findings mean in a real-world context?
Offer 2-3 policy recommendations or actionable insights based on the results.

Tips:

- Keep text minimal and focused.
- Use visuals to highlight key points and keep the audience engaged.
- Stick to a 1-minute per slide rule to keep within the 5-minute limit.
- Use Google Slides to prepare your slides.

4 Instructions for submitting the report and slides

The deadline for the final report and presentation slides submission is today at 17:00. You should do so by sending the files to juryeg@vsae.nl. Title this email “Final Case B - [University Name]”.

5 Guidelines for using ChatGPT and other AI tools

- Allowed Usage:
ChatGPT can be used to assist with coding tasks, including debugging, optimizing, and understanding programming concepts. You may use ChatGPT to generate code snippets, explore different coding approaches, and troubleshoot errors.
- Not Allowed Usage:
ChatGPT cannot be used to write your report, including explanations, analysis, or conclusions. All written content in your report must be your own work, based on your understanding and research.

6 Feature names and descriptions

`begin_date` - Begin date of redispach
`end_date` - End date of redispach
`operator` - Operator
`direction` - Direction
`mean_load` - Average load
`max_load` - Maximum load
`total_load` - Total load
`n_concurrent_redispatches` - Number of concurrent redispatches
`brent_oil_price_open` - Opening price of Brent oil, USD/Bbl
`brent_oil_price_close` - Closing price of Brent oil, USD/Bbl
`brent_oil_price_high` - Highest price of Brent oil during a trade session, USD/Bbl
`brent_oil_price_low` - Lowest price of Brent oil during a trade session, USD/Bbl
`carbon_emissions_futures_price` - Clothing price of carbon emission futures, USD/K
`carbon_emissions_futures_open` - Opening price of carbon emission futures, USD/K
`carbon_emissions_futures_high` - Highest price of carbon emission futures during a trade session, USD/K
`carbon_emissions_futures_low` - Lowest price of carbon emission futures during a trade session, USD/K
`carbon_emissions_futures_vol` - Trade volume of carbon emission futures pro trade session, in thousands of units
`electricity_day_ahead_price_ch` - Day-ahead price of electricity in Switzerland, USD/MWh
`electricity_day_ahead_price_cz` - Day-ahead price of electricity in Chech Republic, USD/MWh,
`electricity_day_ahead_price_be` - Day-ahead price of electricity in Belgium, USD/MWh

electricity_day_ahead_price_nl - Day-ahead price of electricity in the Netherlands, USD/MWh
electricity_day_ahead_price_pl - Day-ahead price of electricity in Poland, USD/MWh
electricity_day_ahead_price_at - Day-ahead price of electricity in Austria, USD/MWh
electricity_day_ahead_price_de_lu - Day-ahead price of electricity in Germany and Luxembourg, USD/MWh
natural_gas_price_open - Opening price of natural gas, USD/MMBtu
natural_gas_price_close - Closing price of natural gas, USD/MMBtu
natural_gas_price_high - Highest price of natural gas during a trade session, USD/MMBtu
natural_gas_price_low - Lowest price of natural gas during a trade session, USD/MMBtu
sunshine_duration - Average sunshine duration per TSO per hour, in minutes
air_temperature - Average air temperature at a height of 2 meters per TSO per hour, in Celcius
wind_speed - Average wind speed per TSO per hour, in m/s
day - Day
hour - Hour
month - Month
is_workday - Workday dummy
is_weekend - Weekend dummy
is_holiday - Holiday dummy
consumption_forecast_grid_load_mwh - Day-ahead forecast of grid load for Germany per hour, MWh
consumption_forecast_residual_load_mwh - Day-ahead forecast of residual load for Germany per hour, MWh
day_ahead_auktion - Day-ahead auction price of CO2 emission certificate, USD/tCO2
co2_emissions_certificate_auction_de - Price of CO2 emission certificate in Germany, USD/tCO2
co2_emissions_certificate_auction_eu - Price of CO2 emission certificate in EU, USD/tCO2
production_forecast_total_mwh - Day-ahead forecast of total electricity production in Germany per hour, MWh
production_forecast_photovoltaics_and_wid_mwh - Day-ahead forecast of wind and photovoltaic electricity generation in Germany per hour, MWh
production_forecast_wind_offshore_mwh - Day-ahead forecast of wind and photovoltaic electricity generation in Germany per hour, MWh
production_forecast_wind_onshore_mwh - Day-ahead forecast of wind onshore electricity generation in Germany per hour, MWh
production_forecast_photovoltaics_mwh - Day-ahead forecast of photovoltaic electricity generation in Germany per hour, MWh
production_forecast_other_mwh - Day-ahead forecast of electricity generated from conventional sources in Germany per hour, MWh

mean_sunshine_duration_bayern - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_bayern - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_bayern - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_bremen - - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_bremen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_bremen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_hessen - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_hessen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_hessen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_niedersachsen - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_niedersachsen - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_niedersachsen - Maximum sunshine duration in the German federal state per hour, in minutes
mean_sunshine_duration_schleswig_holstein - Average sunshine duration in the German federal state per hour, in minutes
min_sunshine_duration_schleswig_holstein - Minimum sunshine duration in the German federal state per hour, in minutes
max_sunshine_duration_schleswig_holstein - Maximum sunshine duration in the German federal state per hour, in minutes
mean_wind_velocity_bayern - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_bayern - Minimum wind speed per hour in the German federal state, in m/s
max_wind_velocity_bayern - Minimum wind speed per hour in the German federal state, in m/s
mean_wind_velocity_bremen - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_bremen - Minimum wind speed per hour in the German federal state, in m/s
max_wind_velocity_bremen - Minimum wind speed per hour in the German federal state, in m/s
mean_wind_velocity_hessen - Maximum sunshine duration in the German federal state per hour, in minutes
min_wind_velocity_hessen - Minimum wind speed per hour in the German federal state, in m/s

`max_wind_velocity_hessen` - Minimum wind speed per hour in the German federal state, in m/s
`mean_wind_velocity_niedersachsen` - Maximum sunshine duration in the German federal state per hour, in minutes
`min_wind_velocity_niedersachsen` - Minimum wind speed per hour in the German federal state, in m/s
`max_wind_velocity_niedersachsen` - Minimum wind speed per hour in the German federal state, in m/s
`mean_wind_velocity_schleswig_holstein` - Maximum sunshine duration in the German federal state per hour, in minutes
`min_wind_velocity_schleswig_holstein` - Minimum wind speed per hour in the German federal state, in m/s
`max_wind_velocity_schleswig_holstein` - Minimum wind speed per hour in the German federal state, in m/s

7 Literature

Billault-Chaumartin, Florence, Manuel Eising, and Agustín Motte-Cortés. "Redispatch patterns of energy market and reserve power plants in Germany." 2020 17th International Conference on the European Energy Market (EEM). IEEE, 2020.

Gürses-Tran, Gonca, Hendrik Flamme, and Antonello Monti. "Probabilistic load forecasting for day-ahead congestion mitigation." 2020 International Conference on Probabilistic Methods Applied to Power Systems (PMAPS). IEEE, 2020.

Titz, Maurizio, Sebastian Pütz, and Dirk Witthaut. "Identifying drivers and mitigators for congestion and redispatch in the German electric power system with explainable AI." *Applied Energy* 356 (2024): 122351.

Zhou, Qun, Leigh Tesfatsion, and Chen-Ching Liu. "Short-term congestion forecasting in wholesale power markets." *IEEE Transactions on Power Systems* 26.4 (2011): 2185-2196.